

Sahil Kale

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LinkedIn | Google Scholar | Website

Work experience

Mastercard

Software Engineer II | Mar 2026 - Aug 2026 | Pune, India

- Redesigned the entire data architecture for the Click2Pay and Secure Card-on-File systems, serving over **14 million transactions per month**

Software Engineer I | Jun 2024 - Mar 2026 | Pune, India

- Integrated **AI agents into Apache data flows**, enabling automation and concurrency and resulting in a nearly **35% reduction** in transaction settlement and clearance time
- Supported development of distributed **automated billing batches** handling 1 million+ transactions in 6 global regions

KnowledgeVerse AI

AI Research Consultant | Jun 2024 – Aug 2026 | Atlanta, United States (Remote)

- **Led 5+ research projects** on creating novel LLM benchmarks focused on trustworthiness and explainability of NLP systems in real-world deployment published in notable venues such as **ACL and NAACL**
- Built **over 80 custom agentic AI workflows** to maximize trust and reliability for critical client applications

Stride.ai

Applied Scientist Intern (NLP) | Sep 2023 – Jun 2024 | Bangalore, India

- Led and architected a **scalable production NLP system (processes 40,000 words in under 2 minutes)** for generating domain-specific FAQs across 17 fields that can use domain jargon and specific constructs

Barclays

Software Engineer Intern | Jun 2023 – Aug 2023 | Pune, India

- Performed SQL partitioning to **reduce query time to less than 89% of original** for post-trade databases
- Implemented automation and **monitoring of over 90 million transactions recorded on 35 distributed servers**

Education & Training

M.S. Computer Science | University of California, Los Angeles | September 2026 – Dec 2027

- Masters' degree focused on **NLP** and **Artificial Intelligence** at the Henry Samueli School of Engineering

Lisbon Machine Learning Summer School (LxMLS 2025) | Instituto Superior Técnico | July 2025

Grade: 4.0/4.0

- Invited with a full **$1500 Google scholarship** to present novel research on LLM self-knowledge in a PhD school

B.E. Computer Engineering | Pune Institute of Computer Technology | July 2020 – April 2024

Grade: 9.78/10

- Recipient of the Late Ajay Prabhakar Nene Award and fellowship for achieving the **1st rank among 8200 students in the B.E. undergraduate program** in the entirety of the University of Pune

Skills

AI and ML Research | Natural Language Processing | Large Language Models | ML Algorithms | Agentic AI | Model Fine-Tuning | Model Benchmarking | LLM Unlearning | ML Libraries | RAG
Programming Languages | C++ | Python | Unix/Linux
Database Management | SQL | SQL Server | NoSQL
Data Science and Engineering | Apache Hadoop | Apache NiFi | Apache Spark | Big Data | Data Analysis
Web Development | HTML | CSS | FastAPI | Flask
Cloud Computing and Infrastructure | AWS | Microsoft Azure | Microsoft Copilot | Docker | Git | Bash

Selected Publications

ConceptGuard: Benchmarking Context-Sensitive Unlearning in Large Language Models

U&ME Workshop @ ECCV 2026, Malmo - https://arxiv.org/abs/2608.20338

- Contributes a ~5k rows benchmark called ConceptGuard for safety-aware LLM unlearning, where forget and retain sets are explicitly complementary in concept usage to maximize contextual separation and promote safer behavior

Local Information Access in Marathi: Evaluating LLM-Native Web Retrieval in a Low-Resource Environment

ACM SIGIR Conference 2026, Melbourne - https://dl.acm.org/doi/abs/10.1145/3805712.3808534

- Identifies challenges in AI-backed information retrieval for under-represented languages isolating failure modes in translated global queries and natively crafted local information needs

Designing Policy with Last-Mile Stakeholders: Connecting Ground-Level Farmer Insights to Indian Agrarian Policymakers with NLP

ACM DIS Conference 2026, Singapore - https://dl.acm.org/doi/abs/10.1145/3800645.3812895

- Explores how NLP and voice-based AI can connect ground-level farmer insights with policymakers, enabling last-mile perspectives to inform agricultural policy design and formulation

KnowRL: Teaching Language Models to Know What They Know

ArXiv Preprint - https://arxiv.org/abs/2510.11407

- Presents a framework that strengthens a model's internal understanding of its own feasibility boundaries such that with just a small seed set and no external supervision, our method drove gains of 28% in accuracy and 12% in F1

Lie to Me: Knowledge Graphs for Robust Hallucination Self-Detection in LLMs

ICAART Conference 2026, Marbella - https://www.scitepress.org/PublicationsDetail.aspx?ID=RcMLaWCJ6gk=&t=1

- Demonstrates that structuring LLM outputs as knowledge graphs significantly improves hallucination self-detection, achieving up to 16% higher accuracy and 20% better F1-score over existing methods

Mirage of Mastery: Memorization Tricks LLMs into Artificially Inflated Self-Knowledge

AAAI IASEAI Conference 2026, Paris - https://ojs.aaai.org/index.php/IASEAI/article/view/43032

- Shows a noteworthy problem in LLM generalization: models draw confidence from memorized solutions to infer a higher self-knowledge about their reasoning ability, seen as an over 45% inconsistency in feasibility assessments

Line of Duty: Evaluating LLM Self-Knowledge via Consistency in Feasibility Boundaries

TrustNLP Workshop @ NAACL 2025, Albuquerque - https://aclanthology.org/2025.trustnlp-main.10

- Obtains intrinsic insights into different types of LLM self-knowledge with a novel methodology and found that even frontier models like GPT-4o and Mistral Large are not sure of their capabilities more than 80% of the time!

TeXpert: A Multi-Level Benchmark for Evaluating LaTeX Code Generation by LLMs

5th Workshop on Scholarly Document Processing @ ACL 2025, Vienna - https://aclanthology.org/2025.sdp-1.2

- Proposes TeXpert, our dataset with natural language prompts for generating LaTeX code focused on components of scientific documents across multiple difficulty levels, showing how poorly LLMs perform as complexity increases

Honours and Awards

Zurich GenAI Awards 2026 - Top 3 | GenAI Zurich | 2026

Awarded as one of the top 3 projects among 70+ submissions in the Impact Achiever category for a solution that uses generative AI to deliver meaningful societal benefits through a jury vote

FAO China Youth Hackathon on Urban Agriculture - Winner | UN FAO & CAAS | 2024

Won first place among 200+ entries from 10 UN member nations for building an efficient AI-powered audio-based data collection and analysis mechanism to document ground-level farmer challenges

Smart India Hackathon - Winner | Government of India | 2023

Achieved first place from 5000+ entries with a $1000 prize for building a "Contact Center Knowledge Management Tool" for the Ministry of Commerce and Industry, Government of India in a 36-hour non-stop hackathon

Languages

English | Marathi | Hindi | German | Japanese

Certifications

Large Language Model Operations (LLMOps) Specialisation | Duke University | 2025

Certification in Building LLM-Powered Applications | Weights and Biases | 2024

Specialisation in Natural Language Processing | Great Learning | 2023

Machine Learning with Python: Gradient Boosting and Advanced Techniques | Jovian AI | 2022

Projects

Sampark Sarita: AI Knowledge Management - https://github.com/Sahil-R-Kale/Sampark-Sarita-SIH

- An **agentic AI-powered question-answering tool** built for the Ministry of Commerce and Industry, India
- Reduced **query resolution time by 32.5%** with a dashboard that guides agents to the perfect series of questions and follow-up answers with real-time suggestions and multilingual voice integration

ClashFree: AI-Powered Scheduling Assistant - https://github.com/Sahil-R-Kale/Clash-Free

- An automatic, user-friendly schedule builder equipped to solve **scheduling conflicts up to 1-minute granularity**
- Provides **automatic suggestions using a genetic AI algorithm**, drag-and-drop UI for human-in-the-loop validations and a scalable database integration for future reference

Finterest: AI-Powered Financial Planning - https://github.com/Sahil-R-Kale/Finterest.git

- A personal financial planner providing AI-powered personalised **portfolio diversification based on 5 parameters of risks**, along with short and long-term investment goal planning
- Provides a precise and customised report using cloud services suggesting **10+ diversified investment options**

Volunteering

Coding Tutor | Young Thinks Foundation | Mar 2023 – May 2023

Helped underprivileged students in rural India improve their analytical and technical skills in Python and C++ through practical mentoring sessions

English Tutor | Read-a-Story Foundation | Sep 2022 – Feb 2023

Conducted 1:1 biweekly tutoring to improve reading and spoken English in the remote regions of India with an outcome of improved word retention by 42%